Dimitrios Athinaios

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Education

UCL School of Management, London, UK - Current Grade: *75% Distinction* **Sep/2025 – Sep/2026**

MSc Business Analytics

Relevant Modules: Data Engineering (MLOps, Docker, AWS, Git, SQL), Operational Analytics Machine Learning (Linear Algebra, Multivariate Calculus, PCA. Stats), Predictive Analytics.

Henley Business School, University of Reading, UK - Grade: *First Class Honours* **Sep/2022 – Jun/2025**

BSc Business Management(Data Analytics)

Relevant Modules: Data Analytics, Machine Learning Introduction, Information Systems

Work Experience

Intelmatix (Dissertation Placement), London, UK – Geospatial Data Scientist. **April/2026 – Sep/2026**

- Engineering a spatial network of 1M+ POIs using Python and NetworkX to map urban connectivity and functional dependencies.
- Architecting serverless pipelines via AWS (S3, Glue, Athena) to automate the processing of high-dimensional geospatial datasets at scale.
- Analyzing ecosystem-level insights using clustering and co-occurrence matrices to drive strategic market location and urban planning decisions.

Globassure, Athens, Greece – Data Scientist. **May/2024 – August/2024**

- Built a data scraping pipeline to identify and extract potential client leads, securing 7 start-up businesses for employee insurance coverage through targeted outreach.
- Cleaned and processed 120,000+ records, performed exploratory data analysis (EDA), and identified key data relationships to surface actionable business insights.

Projects

London Synergy Index – Predictive Site Selection – Python, XGBoost, H3, NetworkX, Streamlit – [GitHub]

- Architected an end-to-end geospatial pipeline for 55,000 H3 hexagons, integrating LandScan satellite rasters, ONS Census data, and OSM POIs enriched with graph-centrality and competition-density features.
- Modeled underserved retail locations using spatially cross-validated XGBoost classifiers; applied Burt's Structural Hole Theory to identify market gaps from model false positives, deployed via a Streamlit dashboard with 3D Pydeck visualizations.

Geospatial London Housing & Crime Analysis – Python, Scikit-learn, XGBoost – [GitHub]

- Developed a Spatio-Temporal pipeline merging 1M+ records to engineer a predictive "Opportunity Index," transforming raw crime and housing data into an actionable investment strategy.
- Optimized an XGBoost regressor via Optuna, achieving an R^2 of 0.92; outperformed KNN and Polynomial baselines by effectively capturing complex, non-linear location-safety interactions.

RiskTerrain – Geospatial S&P 500 Risk Intelligence – Python, LangGraph, SurrealDB, Three.js – [GitHub]

- Engineered a 6-node LangGraph pipeline that ingests live geopolitical events (USGS, NewsAPI), performs multi-hop supply chain graph traversal across 154 companies in SurrealDB, and scores cascading risk exposure in under 8 seconds.
- Implemented hybrid graph-vector retrieval combining ~90 weighted RELATE edges with sentence-transformer embeddings for historical event similarity; containerized via Docker Compose and deployed to GitHub Pages.

Data Engineering Pipeline – Python, Pyspark, MongoDB – [GitHub]

- Architected a centralized Star Schema Lakehouse to unify semi-structured NoSQL and CSV datasets; utilized PyMongo and Pandas to flatten nested JSON and engineer derived geographical dimensions.
- Streamlined analytical query performance and storage efficiency by implementing columnar Parquet formats and integrating DuckDB for high-speed, in-process SQL aggregations.

Dataportfolio.co.uk – **Full Stack (Backend APIs, Frontend, Database)** – [Live Site]

- Launched a production-ready SaaS platform that automates portfolio creation for data professionals, featuring a "coding-less" GUI to manage backend data persistence and dynamic metadata updates.

Professional Skills

Programming: Python (Pandas, Numpy, Scikit-learn), SQL, R, JavaScript.
Machine Learning: XGBoost, PyTorch, TensorFlow, LLMs (LangChain, Ollama), Optuna, SHAP, Graph Theory.
Data Engineering: AWS (S3, Glue, Athena), Databricks, PySpark, Docker, Kubernetes, Git, DuckDB, Star Schema.
Geospatial: H3 Indexing, Pydeck, LandScan Raster, OSM Data, Spatio-Temporal Analytics, Graph Centrality.
Spoken Languages: Greek (Native), English (C2), Spanish (B1).
Certifications: Mathematics for Machine Learning Specialisation – Imperial College | Google Advanced Data Analytics | Microsoft Power BI Desktop | IBM Data Analyst |

Extra-Curricular Activities

Co-Founder/President of Data Analytics Society, University of Reading **2023 – 2025**

- Led a student community, organizing technical workshops and industry networking events.

Interests and Personal Development : Enthusiastic about technology and martial arts (Karate Black Belt 1-dan, Kickboxing), fitness (gym, running, calisthenics), music, and competitive pursuits. Committed to personal growth in data science, machine learning, LLMs, AI Engineering, and entrepreneurship, guided by Kaizen principles.